Number — Negative integers · Prime factors · Indices · Priority of operations

No calculator. Show your working. ★ = challenge question.

A · NEGATIVE INTEGERS

1 The temperature in Oslo was -6 °C at 7 am and 3 °C at 2 pm.

a Work out the rise in temperature between 7 am and 2 pm.

b By midnight the temperature had fallen by 11 °C compared with 2 pm. What was the temperature at midnight?

c Work out $-3 - 7$.

2 Work out

a -4×6 **b** $(-3) \times (-7)$ **c** $45 \div (-9)$ **d** $-36 \div (-4)$

e $-2 \times (-3) \times (-5)$

3 Substitute the values into each formula and work out the answer.

a $v = u + at$ when $u = -5$, $a = 3$ and $t = 4$

b $A = 2b - c^2$ when $b = -3$ and $c = -4$

4 Maya says that $(-5)^2 = -25$. Explain her mistake and write the correct value.

B · PRIME FACTOR DECOMPOSITION, HCF AND LCM

5 Write each number as a product of its prime factors, using index notation.

a 84 **b** 360

6 Use your answers to question 5 to find

a the HCF of 84 and 360 **b** the LCM of 84 and 360

7 Two buses leave a station at 9:00. Bus A leaves every 12 minutes and bus B every 18 minutes. At what time do they next leave together?

C · INDICES

8 Write each calculation as a single power.

a $7^4 \times 7^3$ **b** $5^9 \div 5^4$ **c** $(2^3)^4$ **d** $3^6 \times 3 \div 3^2$ **e** $10^8 \div 10^8$

9 Evaluate $\dfrac{2^5 \times 2^3}{2^6}$

10 Which calculation does **not** give the same answer as the others? Explain.

A $(2 \times 3)^2$ **B** 2×3^2 **C** $2^2 \times 3^2$ **D** 6^2

D · PRIORITY OF OPERATIONS

11 Work out

a $4 + 3 \times 2^2$ **b** $(4 + 3) \times 2^2$ **c** $20 - \sqrt{36} \div 2$ **d** $(10 - 2^3)^2 \times 3$

e $\sqrt[3]{27} + 5 \times 2$ **f** $100 \div (3 + 2)^2$

12 Insert one pair of brackets to make this calculation correct: $3 + 5 \times 2 - 1 = 15$

13 Evaluate $\dfrac{5 \times 2^4}{4 \times 10}$ and $\dfrac{(7 - 4)^2}{3^2 - 6}$

Check: a sign for every negative answer · index laws only work for the *same* base · brackets → indices → ÷ × → + −.